# Solar O appendix

Sun · O · downloadable appendix

| | |
|---|---|
| **Document** | Solar O appendix |
| **Subject** | Solar photospheric abundance appendix · O |
| **Author** | Ryan Schmitt |
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Solar oxygen: 8.751 ± 0.077

Asplund, Amarsi & Grevesse 2021: 8.69 ± 0.04 · this measurement +0.061 dex

HARPS has no Reference Grade product for this element, so this is our best Reference Grade product from CRIRES+ or Kitt Peak, chosen the same way: the most complete treatment, then the most lines. Products and engines are not averaged together.

red-optical: 8.751 ± 0.077 · Kitt Peak (Kurucz 2005) · red-optical · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)

## Highlighted Products

### VIS

**8.934 ± 0.395**

±0.309 stat · ±0.246 syst

[O I] 6300 Å forbidden line (blended with Ni I) · 1D LTE · HARPS · Codex Grade

### red-optical

**8.751 ± 0.077**

±0.029 stat · ±0.071 syst

O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (Kurucz 2005) · Reference Grade

### NIR

**8.651 ± 0.103**

±0.074 stat · ±0.072 syst

O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · IAG FTS · Reference Grade

## Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

| Grade | What it means | This element |
|---|---|---|
| Reference Grade | lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used | 18 products -- O I · Asplund et al. 2021 lines |

| Grade | What it means | This element |
|---|---|---|
| Codex Grade | lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep | 10 products -- O I · Codex-graded lines; [O I] 6300 Å forbidden line (blended with Ni I) |
| Deep Grade | the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere | none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines) |

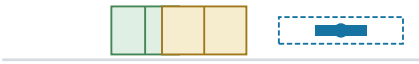
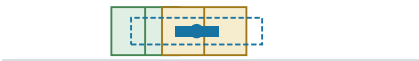
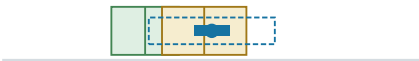
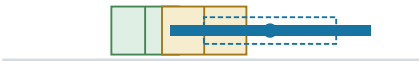
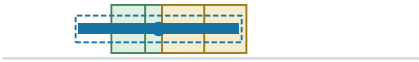
# Error-bar Forest Plot

## VIS — HARPS
## A(O) dex; axis 8.60–9.27

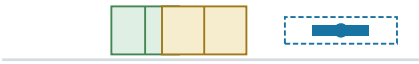
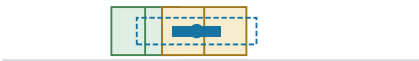
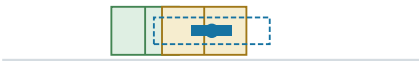
| | | |
|---|---|---|
| [O I] 6300 Å forbidden line (blended with Ni I) · 1D LTE<br>Codex Grade · 1 line |  | 8.934<br>±0.309 stat ±0.246 syst |

VIS axis: 8.60 · 8.76 · 8.93 · 9.10 · 9.27 A(O) dex

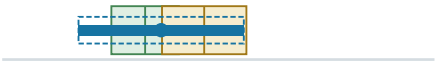
## red-optical — IAG FTS
## A(O) dex; axis 8.52–9.03

| | | |
|---|---|---|
| O I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 4 lines |  | 8.922<br>±0.030 stat ±0.074 syst |
| O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 4 lines |  | 8.751<br>±0.026 stat ±0.078 syst |
| O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 4 lines |  | 8.769<br>±0.021 stat ±0.075 syst |
| O I · Codex-graded lines · 1D LTE<br>Codex Grade · 2 lines |  | 8.838<br>±0.119 stat ±0.078 syst |
| O I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 8.677<br>±0.108 stat ±0.102 syst |
| O I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 2 lines |  | 8.706<br>±0.095 stat ±0.098 syst |

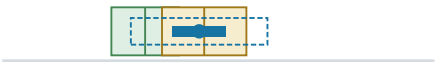
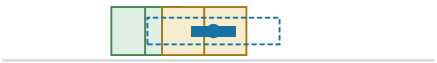
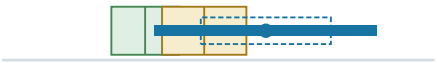
## red-optical — Kitt Peak (Kurucz 2005)
## A(O) dex; axis 8.52–9.03

| | | |
|---|---|---|
| O I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 4 lines |  | 8.922<br>±0.034 stat ±0.066 syst |
| O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 4 lines |  | 8.751<br>±0.029 stat ±0.071 syst |
| O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 4 lines |  | 8.769<br>±0.024 stat ±0.069 syst |
| O I · Codex-graded lines · 1D LTE<br>Codex Grade · 2 lines |  | 8.841<br>±0.122 stat ±0.078 syst |

## red-optical — Kitt Peak (Kurucz 2005)
## A(O) dex; axis 8.52–9.03

| | | |
|---|---|---|
| O I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 8.680<br>±0.111 stat ±0.100 syst |
| O I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 2 lines |  | 8.709<br>±0.098 stat ±0.098 syst |

## red-optical — Kitt Peak (1984, molecfit)
## A(O) dex; axis 8.52–9.03

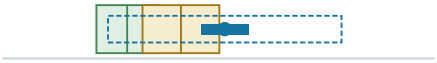
| | | |
|---|---|---|
| O I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 4 lines |  | 8.925<br>±0.036 stat ±0.078 syst |
| O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 4 lines |  | 8.754<br>±0.032 stat ±0.081 syst |
| O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 4 lines |  | 8.771<br>±0.026 stat ±0.078 syst |
| O I · Codex-graded lines · 1D LTE<br>Codex Grade · 2 lines |  | 8.833<br>±0.132 stat ±0.077 syst |
| O I · Codex-graded lines · 3D NLTE (Amarsi)<br>Codex Grade · 2 lines |  | 8.672<br>±0.121 stat ±0.100 syst |
| O I · Codex-graded lines · 1D NLTE (Bergemann)<br>Codex Grade · 2 lines |  | 8.701<br>±0.107 stat ±0.097 syst |

red-optical axis: 8.52 · 8.65 · 8.78 · 8.90 · 9.03 A(O) dex

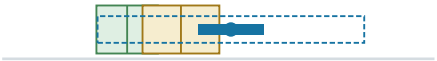
## NIR — IAG FTS
## A(O) dex; axis 8.53–9.09

| | | |
|---|---|---|
| O I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 3 lines |  | 8.742<br>±0.075 stat ±0.063 syst |
| O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 3 lines |  | 8.651<br>±0.074 stat ±0.072 syst |
| O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 3 lines |  | 8.689<br>±0.071 stat ±0.067 syst |

## NIR — Kitt Peak (Kurucz 2005)
## A(O) dex; axis 8.53–9.09

| | | |
|---|---|---|
| O I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 3 lines |  | 8.910<br>±0.031 stat ±0.150 syst |
| O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 3 lines |  | 8.817<br>±0.031 stat ±0.152 syst |
| O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 3 lines |  | 8.848<br>±0.030 stat ±0.146 syst |

**NIR — Kitt Peak (1984, molecfit)**
**A(O) dex; axis 8.53–9.09**

| | | |
|---|---|---|
| O I · Asplund et al. 2021 lines · 1D LTE<br>Reference Grade · 3 lines |  | 8.825<br>±0.043 stat ±0.173 syst |
| O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)<br>Reference Grade · 3 lines |  | 8.733<br>±0.042 stat ±0.176 syst |
| O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)<br>Reference Grade · 3 lines |  | 8.768<br>±0.041 stat ±0.168 syst |

NIR axis: 8.53 · 8.67 · 8.81 · 8.95 · 9.09 A(O) dex

**How to read this forest.** Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

## How this was measured

Every oxygen value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the oxygen abundance as the free parameter. The synthesis includes the atomic lines and the molecular bands (CN, CH, $C_2$, OH and others) in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, blends and coupling to the other C/N/O abundances, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

| Band | What was measured | Solar spectra | Treatments | Lines | Result A(O) | Largest uncertainty terms (dex) |
|---|---|---|---|---|---|---|
| VIS | [O I] 6300 Å forbidden line (blended with Ni I) | HARPS | 1D-LTE | 1 | 8.934 (1D-LTE) | fit / line-to-line scatter 0.31, continuum placement 0.17 |
| red-optical | O I · Codex-graded lines | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) | 1D-LTE, 1D-NLTE, 3D-NLTE | 2 | 8.672–8.680 (3D-NLTE) | fit / line-to-line scatter 0.11, NLTE / 3D correction 0.07 |
| red-optical | O I · Asplund et al. 2021 lines | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) | 1D-LTE, 1D-NLTE, 3D-NLTE | 4 | 8.751–8.754 (3D-NLTE) | continuum placement 0.05, model atmosphere 0.03 |

| Band | What was measured | Solar spectra | Treatments | Lines | Result A(O) | Largest uncertainty terms (dex) |
|---|---|---|---|---|---|---|
| NIR | O I · Asplund et al. 2021 lines | IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005) | 1D-LTE, 1D-NLTE, 3D-NLTE | 3 | 8.651–8.817 (3D-NLTE) | fit / line-to-line scatter 0.07, telluric residual 0.05 |

## Solar oxygen context

Oxygen is traced by forbidden [O I], permitted O I and molecular OH indicators where available. Blends, departures from LTE and atmospheric structure affect these families differently, so their results retain separate identities.

C/O indicator strategy · Methodology

## Open provenance decisions

O I 615.8nm:  DISAGREES — reported to Ryan, not resolved. . Store log gf -0.409 (); summed pull -0.2966 (B+).

O I 630.0nm:  DISAGREES — reported to Ryan, not resolved.  store claims A, pull publishes B+. Store log gf -9.717 (A); summed pull -9.7744 (B+).

OPEN. Tachiev and NIST differ systematically by -0.016 dex on the 777 triplet and -0.005 dex on 8446. AGSS21 names Tachiev for N i only and states no O i gf source. Both values are reported in oi_tachiev_vs_nist.csv; NIST is what the store already carried and this script does not change it. Ryan decides.

Adjudication record · Indicator gf checks

## Download the evidence

Download PDF — Solar O appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

## Reproducibility

Generated from the pinned science snapshot. Source commit 27afcfcba7db6b4782ca756c1131fb9ddaa8dc3c
Feed O.json v1.102 · updated 2026-10-10T15:46:17Z
Generated 2026-10-28T11:24:25-06:00 (source commit timestamp)
Feed SHA-256 ca940572b4a40d4623022071b580b1999bae222cef798f2628c631b0ef01af90
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

## References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

**Solar abundance and indicator analyses.** Solar O atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [1], [2].

**Atomic data and gf authorities.** The recorded O I/O II base pedigree is Wiese–Fuhr–Deters / Opacity Project theory. Per-transition exceptions and the open O I source decisions remain explicit. Sources: [3], [4], [5].

**Atomic data and gf authorities.** O I triplet source record: NIST — Tachiev NOT adopted; OPEN — under review Sources: [3], [6].

**Model and grid sources.** Registered O I line/grid lineage: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — line-by-line 3D-NLTE grids built on the Amarsi+2018 O I model atom. The current feed treatment remains authoritative; a grid reference does not imply a correction was applied. Sources: [7], [8].

**Atomic data and gf authorities.** Atomic O I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4], [5].

**Model and grid sources.** Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [9].

**Model and grid sources.** Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 3D non-LTE - 1D LTE on STAGGER (table2 C I / table3 O I); PER-LINE additive correction, grid lines only (match <= 0.05 A) Sources: [7], [8].

**Model and grid sources.** Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 1D non-LTE - 1D LTE (table5 C I / table6 O I); PER-LINE additive correction, grid lines only (match <= 0.05 A) Sources: [7], [8].

**Special diagnostics.** The synthesis sidecar names Johansson’s Ni I blend data, Storey & Zeippen’s forbidden-line gf lineage and Caffau’s CO5BOLD comparison anchor. The feed entry remains 1D-LTE and quarantined; the sidecar’s cited anchor is not promoted to a measured abundance. Sources: [10], [11], [12].

Product-to-source index

NIR · IAG FTS · O I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] O-products-2f26fcc4de61

NIR · IAG FTS · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-ee093b5cd43b

NIR · IAG FTS · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-dcccc1bf145a

NIR · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] O-products-7570ea595567

NIR · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-34b4e514053b

NIR · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-f76024e80aca

NIR · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] O-products-97dff9c24866

NIR · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-4ce6582718b9

NIR · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-dbd116adb0fe

VIS · HARPS · [O I] 6300 Å forbidden line (blended with Ni I) · 1D LTE · [1], [3], [4], [5], [9], [10], [11], [12] O-products-951619c74bb3

red-optical · IAG FTS · O I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] O-products-400d2fb23109

red-optical · IAG FTS · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-b43c168af108

red-optical · IAG FTS · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-14585017d015

red-optical · IAG FTS · O I · Codex-graded lines · 1D LTE · [1], [3], [4], [5], [9] O-products-4b0ab56d8136

red-optical · IAG FTS · O I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-b35970fb8af7

red-optical · IAG FTS · O I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-9f2918c00938

red-optical · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] O-products-b87159d4fc51

red-optical · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-1a0fed49853d

red-optical · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-30cc32d77032

red-optical · Kitt Peak (Kurucz 2005) · O I · Codex-graded lines · 1D LTE · [1], [3], [4], [5], [9] O-products-5dd7a576c91e

red-optical · Kitt Peak (Kurucz 2005) · O I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-fabf2e672c03

red-optical · Kitt Peak (Kurucz 2005) · O I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-b90f07a8459c

red-optical · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] O-products-192c0fab28e1

red-optical · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-882b5f8ac19f

red-optical · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-6fccf43002bd

red-optical · Kitt Peak (1984, molecfit) · O I · Codex-graded lines · 1D LTE · [1], [3], [4], [5], [9] O-products-e1b890ab2006

red-optical · Kitt Peak (1984, molecfit) · O I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-d115064cbf77

red-optical · Kitt Peak (1984, molecfit) · O I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-0d45f4f81b47

Complete applicable bibliography

## Solar abundance and indicator analyses

**[1]** — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. source record

**[2]** — Amarsi et al. 2021, A&A 656 A113 DOI: 10.1051/0004-6361/202141384. source record

### Atomic data and gf authorities

**[3]** — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. source record

**[4]** — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) source record

**[5]** — Wiese, W. L.; Fuhr, J. R.; Deters, T. M. (1996). Atomic Transition Probabilities of Carbon, Nitrogen, and Oxygen: A Critical Data Compilation. JPCRD Monograph 7. source record

**[6]** — Tachiev & Froese Fischer 2002, A&A 385, 716 DOI: 10.1051/0004-6361:20011816. source record

### Model and grid sources

**[7]** — Amarsi, A. M.; Nissen, P. E.; Skuladottir, A. (2019). Carbon, oxygen, and iron abundances in disk and halo stars. A&A 630, A104 DOI: 10.1051/0004-6361/201936265. source record

**[8]** — Amarsi, Barklem, Asplund, Collet & Zatsarinny 2018, A&A 616, A89 DOI: 10.1051/0004-6361/201832770. source record

**[9]** — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

### Special diagnostics

**[10]** — Johansson, S.; Litzen, U.; Lundberg, H.; Zhang, Z. (2003). Experimental f-Value and Isotopic Structure for the Ni I Line Blended with [O I] at 6300 A. ApJ 584, L107 DOI: 10.1086/374037. source record

**[11]** — Storey, P. J.; Zeippen, C. J. (2000). Theoretical values for the [O III] 5007/4959 line-intensity ratio and homologous cases. MNRAS 312, 813 DOI: 10.1046/j.1365-8711.2000.03184.x. source record

**[12]** — Caffau, E.; Ludwig, H.-G.; Steffen, M.; Livingston, W.; Bonifacio, P.; Malherbe, J.-M.; Doerr, H.-P.; Schmidt, W. (2015). The photospheric solar oxygen project. III. Investigation of the centre-to-limb variation of the 630 nm [O I]–Ni I blend. A&A 579, A88. source record